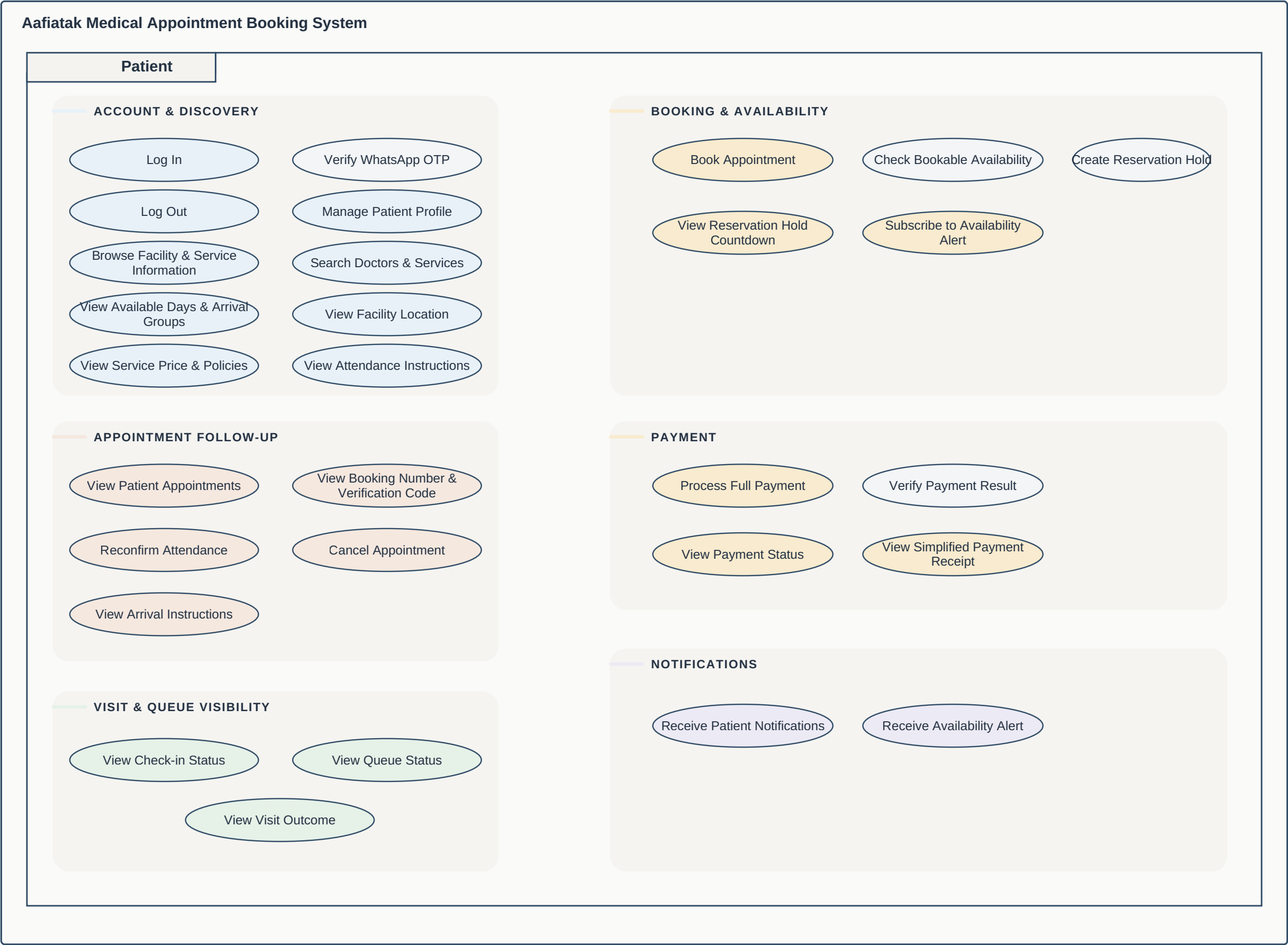
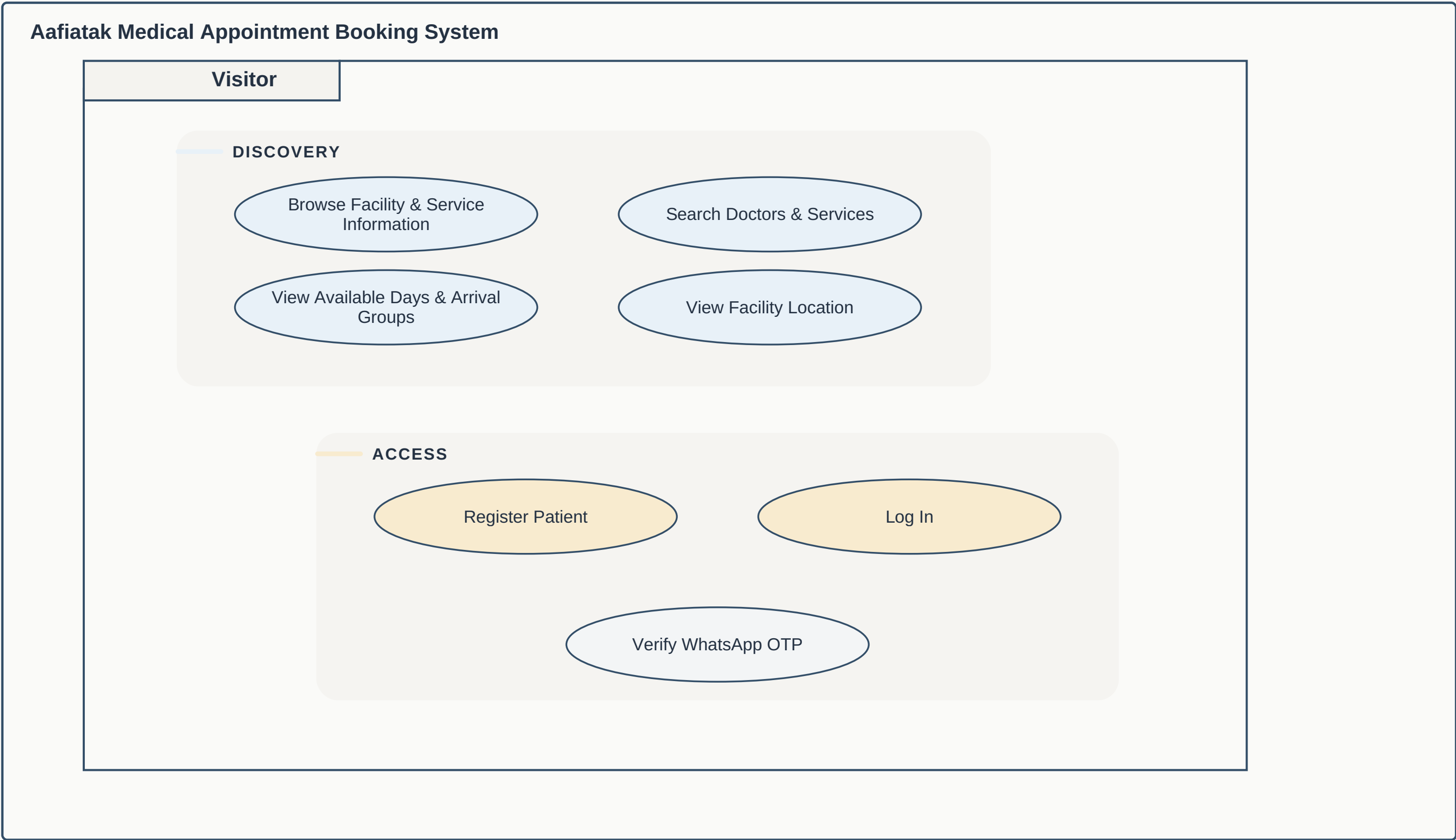


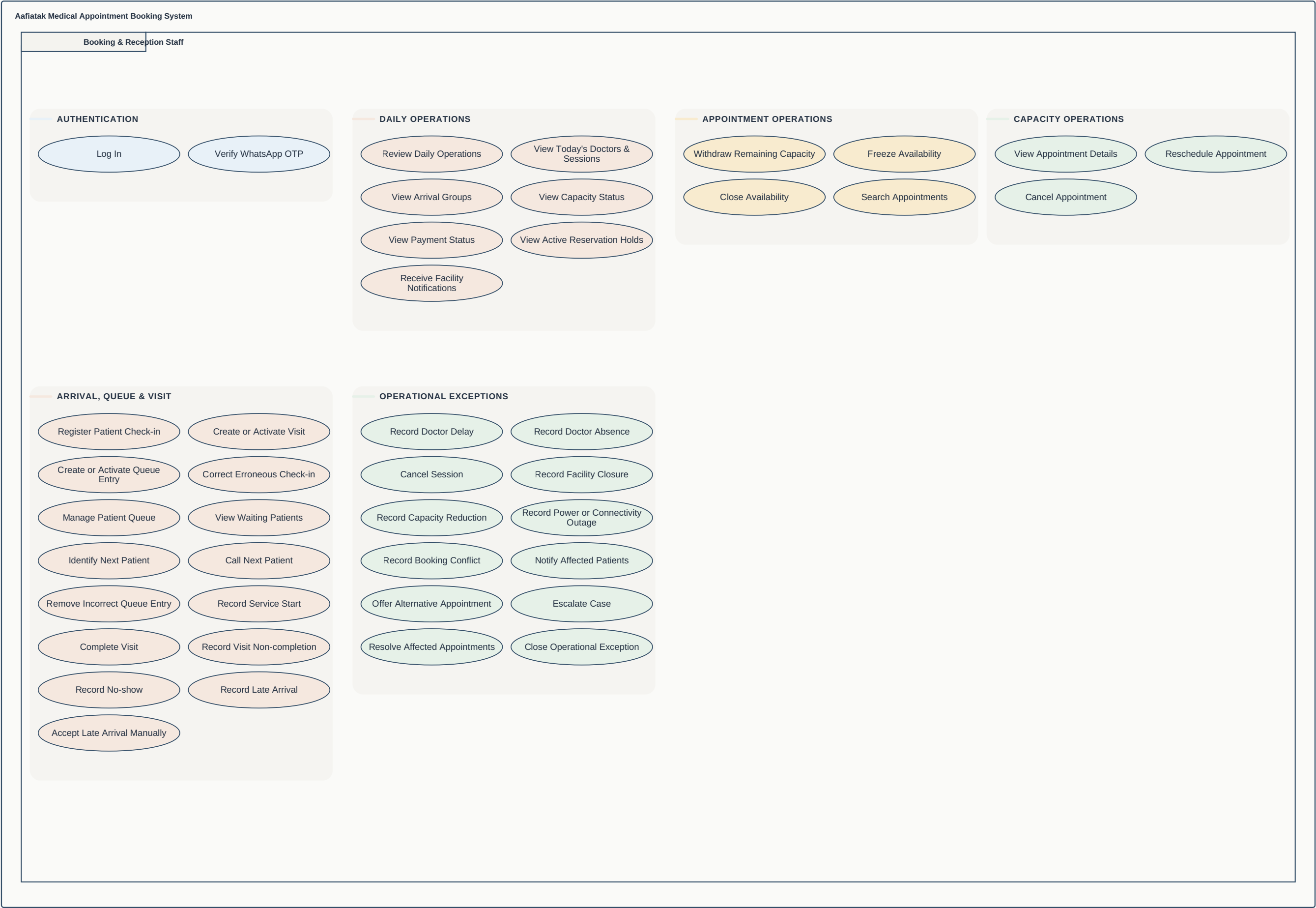
Patient Package Use Case Diagram — Aafiatak Medical Appointment Booking System



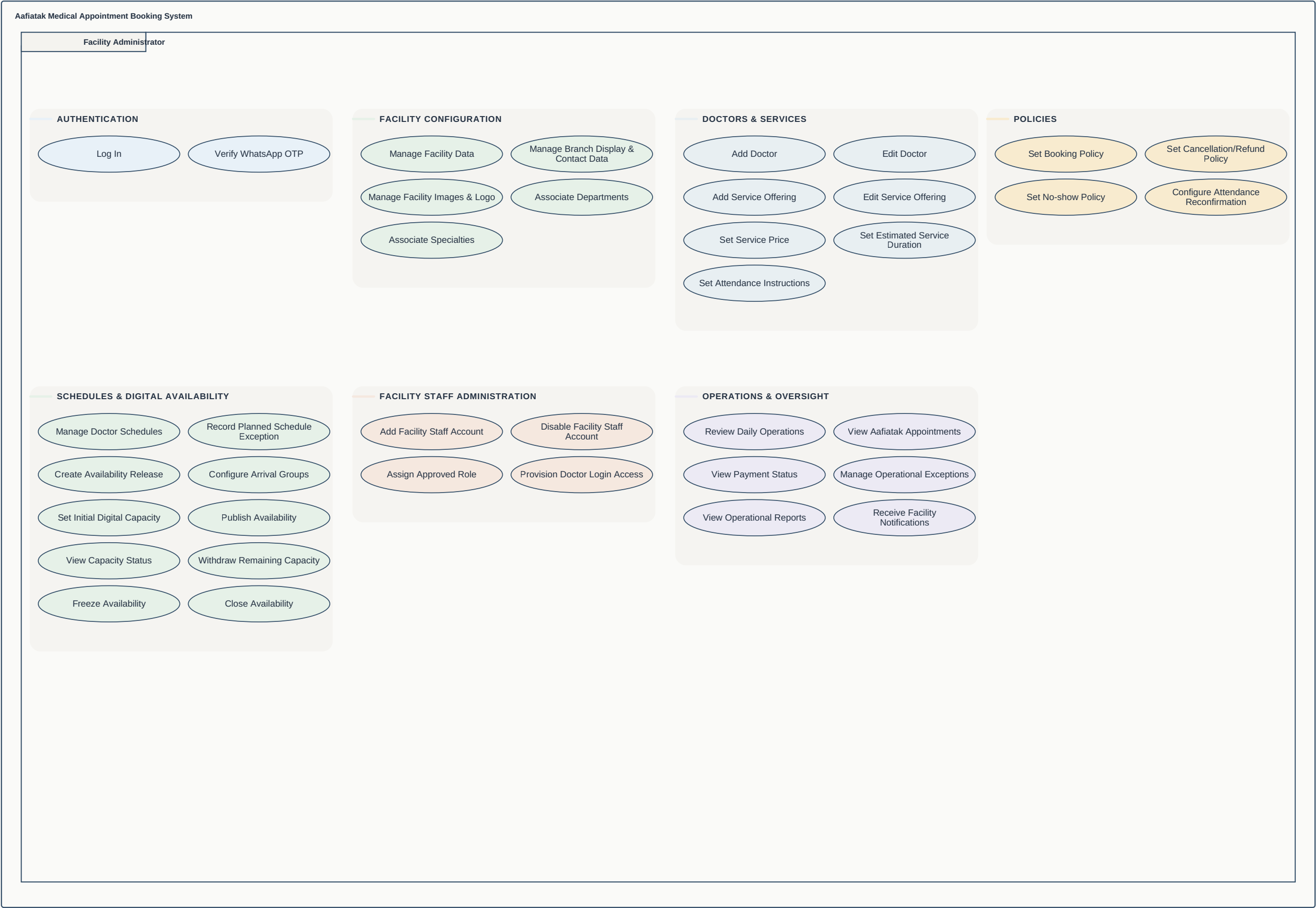
Visitor Package Use Case Diagram — Aafiatak Medical Appointment Booking System



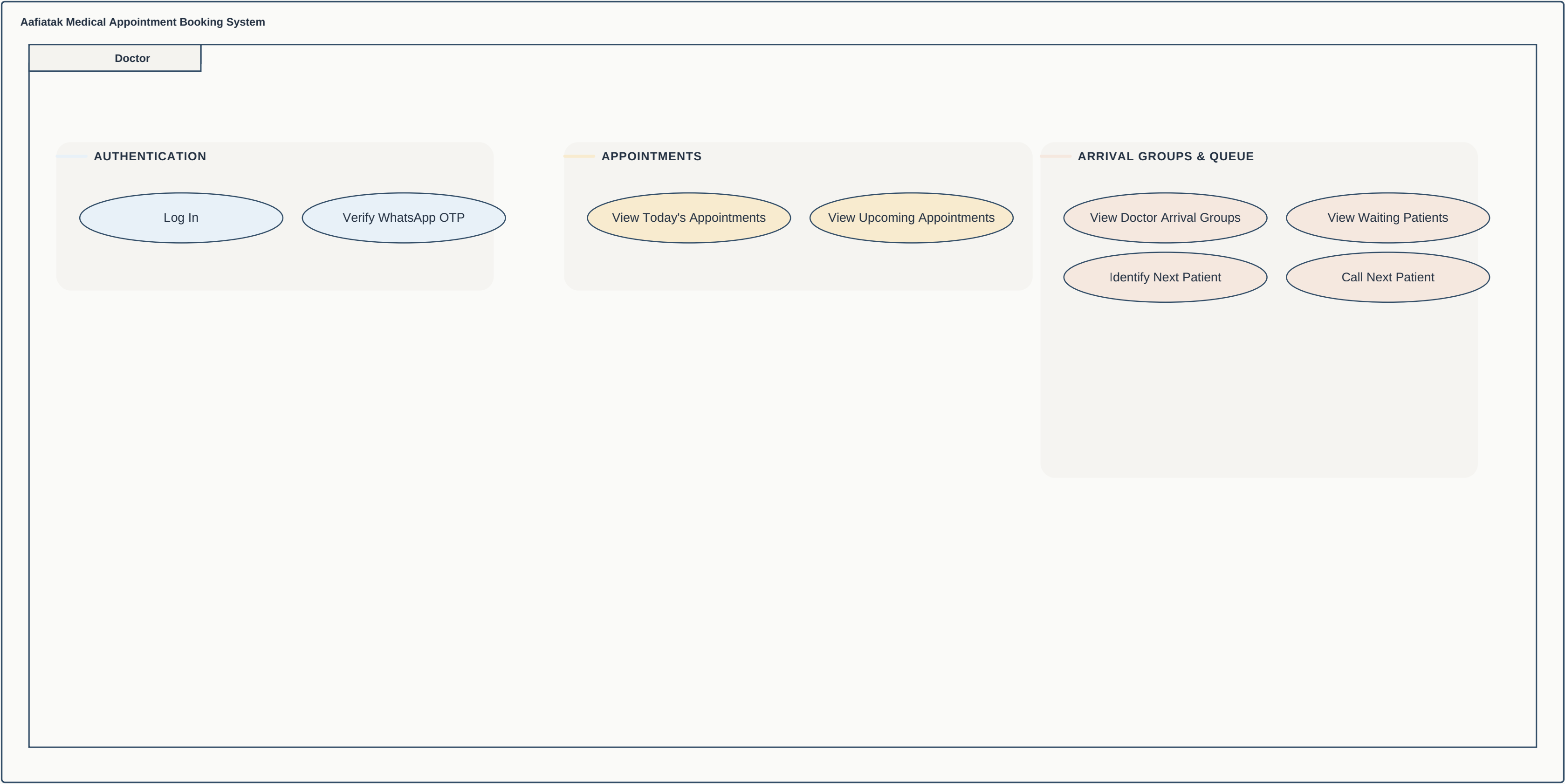
Booking & Reception Staff Package Use Case Diagram



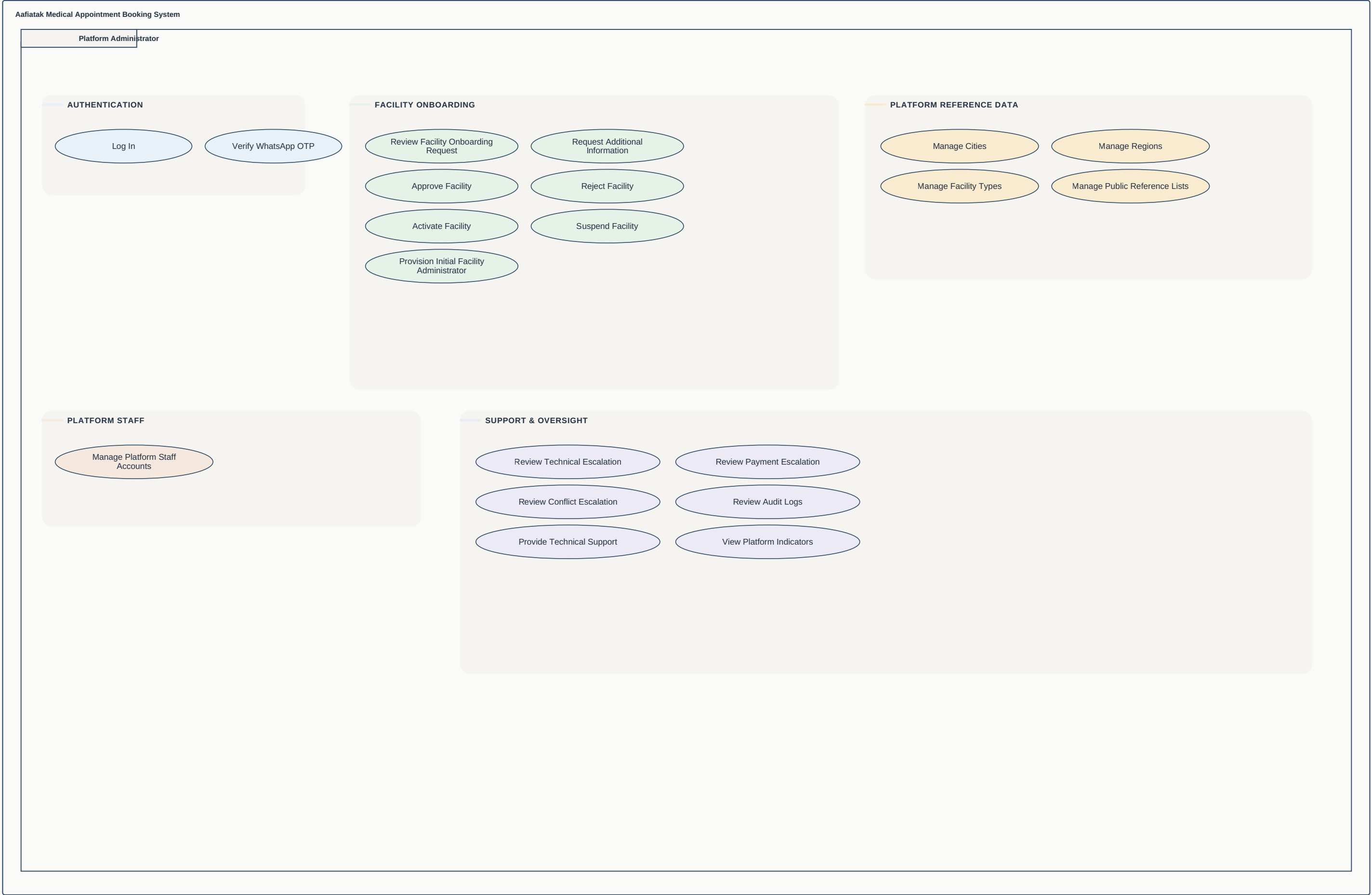
Facility Administrator Package Use Case Diagram



Doctor Package Use Case Diagram — Aafiatak Medical Appointment Booking System



Platform Administrator Package Use Case Diagram — Aafiatak Medical Appointment Booking System



This diagram illustrates a comprehensive system architecture for a healthcare management platform, organized into four main functional areas: Platform & Reference Data, Facility & Medical Offering, Appointment & Payment, and Visit, Queue & Operational Exceptions.

Platform & Reference Data

This section defines the foundational data entities and their relationships:

- User:** Manages user profiles, authentication, and session data. Includes attributes like `userId`, `email`, `password`, and `isActive`.
- Notification:** Tracks system notifications and user alerts. Includes attributes like `notificationId`, `type`, `status`, and `timestamp`.
- AuditRecord:** Records system events and changes for compliance. Includes attributes like `auditId`, `action`, `timestamp`, and `reason`.
- FacilityOnboardingRequest:** Manages the process of adding new facilities. Includes attributes like `requestId`, `status`, `facilityName`, and `onboardingStatus`.
- PlatformRoleAssignment:** Manages user roles and permissions across the platform. Includes attributes like `assignmentId`, `userId`, `role`, and `permissions`.
- Region, City, FacilityType, Facility, FacilityBranch:** Define the geographical and organizational structure of the healthcare network. Includes attributes like `regionId`, `cityId`, `facilityType`, `facilityId`, and `branchId`.
- Department, Specialty:** Define the medical services offered by the facilities. Includes attributes like `departmentId`, `specialtyId`, and `name`.
- Doctor:** Manages medical professionals and their credentials. Includes attributes like `doctorId`, `name`, `email`, `phone`, and `specialty`.
- ServiceOffering:** Defines the specific medical services provided. Includes attributes like `serviceOfferingId`, `serviceName`, `description`, and `duration`.

Facility & Medical Offering

This section details the operational aspects of the healthcare facilities and the services they provide:

- ServiceOffering:** (Detailed view) Includes attributes like `serviceOfferingId`, `serviceName`, `description`, `duration`, `location`, `status`, `availability`, and `pricing`.
- AvailabilityRelease:** Manages the release of service capacity. Includes attributes like `availabilityReleaseId`, `date`, `time`, `status`, and `location`.
- ArrivalGroup:** Tracks groups of patients arriving for a service. Includes attributes like `arrivalGroupId`, `sequence`, `status`, `capacity`, and `location`.
- CapacityWithdrawal:** Manages the withdrawal of service capacity. Includes attributes like `capacityWithdrawalId`, `quantity`, `status`, and `reason`.
- DoctorSchedule:** Tracks the schedule of medical professionals. Includes attributes like `doctorScheduleId`, `date`, `time`, `status`, and `location`.
- ScheduleException:** Manages exceptions to the standard schedule. Includes attributes like `scheduleExceptionId`, `date`, `time`, `status`, and `location`.

Appointment & Payment

This section details the appointment scheduling and payment processing:

- Appointment:** The central entity for scheduling medical appointments. Includes attributes like `appointmentId`, `date`, `time`, `status`, `location`, `doctorId`, and `patientId`.
- AppointmentStatusHistory:** Tracks the status changes of appointments. Includes attributes like `appointmentStatusHistoryId`, `appointmentId`, `status`, and `timestamp`.
- PaymentIntent:** Manages the payment process for appointments. Includes attributes like `paymentIntentId`, `amount`, `status`, `timestamp`, and `location`.
- ReservationHold:** Manages holds on appointments. Includes attributes like `reservationHoldId`, `date`, `time`, `status`, and `location`.
- AvailabilityAlertSubscription:** Manages subscriptions for availability alerts. Includes attributes like `availabilityAlertSubscriptionId`, `targetDate`, `status`, and `location`.

Visit, Queue & Operational Exceptions

This section details the patient visit process and handling of operational exceptions:

- VisitInstance:** Tracks individual patient visits. Includes attributes like `visitInstanceId`, `date`, `time`, `status`, and `location`.
- QueueEntry:** Manages the queue of patients waiting for a service. Includes attributes like `queueEntryId`, `date`, `time`, `status`, and `location`.
- OperationalException:** Manages exceptions to the standard operational process. Includes attributes like `operationalExceptionId`, `date`, `time`, `status`, and `location`.
- OperationalExceptionResolution:** Tracks the resolution of operational exceptions. Includes attributes like `operationalExceptionResolutionId`, `operationalExceptionId`, `status`, and `timestamp`.

The diagram uses a color-coded system to categorize entities: Platform & Reference Data (light blue), Facility & Medical Offering (light green), Appointment & Payment (light orange), and Visit, Queue & Operational Exceptions (light purple). Relationships are indicated by lines connecting the entities, and specific attributes and methods are listed for each entity.